

6. (a) Polymer-based injectable modified release systems come in a range of formats. One format is a polymer microsphere. Describe in point form the advantages and disadvantages of polymer microspheres as a dose form and what each means for the patient. (5 marks) (b) A rival technology to microspheres is 'in situ gelling' polymer systems. In point form briefly describe the attributes of these systems. Is there a clinical benefit to the patient for these systems over microspheres? Why do you think they are of interest? (5 marks)

8. Consider the following five statements, answering whether you (1) agree or disagree and (2) provide a brief reason for your answer. (a) Lyophilization is the only option for preparation of peptides for injectable administration. (2 marks) (b) Protein structure is more complex than peptides, meaning instability is often a greater issue. (2 marks) (c) Antibody-drug conjugates use the natural distribution of a drug to get the antibody to bind to its target on cells to deliver a therapeutic effect. (2 marks) (d) Proteins and peptides are not light sensitive, so packaging to protect the drug from light is never used. (2 marks) (e) Co-solvents are used to dissolve peptides to enable safe injection. (2 marks)